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Global minimum time trajectory planning considering curvature and distance for track racing

Dongmin Zhang✉, Yi Yuan & Futang Zhu

This work focuses on global minimum time trajectory planning for track racing considering curvature and distance based on offline high-precision map of track. Offline mapping of track, post processing and a new gridding method are firstly introduced. Then according to the requirement of quadratic programming method, the sum of discrete squared distance and the sum of discrete squared curvature are derived respectively, which are then combined together by a weight coefficient to constitute the objective function of minimum time trajectory. Drivable curbs, size of vehicle and safe distance are considered in constraint conditions in order to make full use of the width of track. With objective function and constraint conditions, quadratic programming method is used to calculate and iterate a smooth and reasonable trajectory. Petrol and electric racecars, different driver models and different trajectories are simulated in CarSim. Results show that conspicuous improvement can be obtained if using drivable curbs, and the minimum time trajectories for different vehicles and drivers display significant difference.

Keywords Trajectory optimization, Quadratic programming, Curvature and distance, Minimum time, Electric racecars.

The task of trajectory planning is to find a continuous path in space from the initial point to the target point, which should ensure the reliability, safety and comfort of vehicle moving, and satisfy environmental constraints, time constraints and vehicle kinematic constraints. As the rapid development of intelligent assisted driving technology, trajectory planning has been applied to many scenarios, such as changing lanes on highways[1], turning at urban road intersections[2], automatic parking[3], and even automatic driving on racetracks[4]. The trajectory planning algorithms used in different driving scenarios are also different, and widely used trajectory planning algorithms include graph search-based algorithms[5–7], sampling-based algorithms[8–16], optimization-based algorithms[17–19], and deep reinforcement learning method[3,20–22]. Changing lanes, turning, and parking are all routine driving conditions, while racing on racetrack faces more extreme driving conditions. To complete one or more laps in as little time as possible, researchers have proposed many planning methods to improve the lap time based on a reasonable driving trajectory. The methods of track trajectory planning can be divided into two categories according to the planning scope: local planning and global planning.

Local trajectory planning mainly aims is to plan the vehicle's operations (acceleration, braking, and steering) within a fixed time step to find the optimal driving trajectory. The current local trajectory planning methods mainly use control optimization algorithms[23–27]. Funke[28] and Kalaria et al.[29] use local trajectory optimization to modify the global planning trajectory and achieve obstacle avoidance. Evans et al.[30] proposed a local map framework for unmapped tracks by using easily extractable, low-level features to build local maps of the visible region, and its lap time only 3.28% slower than global methods.

Global planning involves finding the complete driving trajectory from the starting point to the endpoint within the boundaries of drivable area, based on a certain objective such as lap time, trajectory geometry, and energy consumption. Some researchers used optimal control theory to optimize trajectory planning with lap time as the objective. Metz et al.[31] transformed trajectory planning into a two-point boundary value problem by establishing an optimization control problem and solving it using a quasi-linearization method. Pagot et al.[32] proposed a nonlinear online MPC optimization control framework for solving the shortest time trajectory of the track, and used this method to test on a 1:8 remote control (RC) vehicle. The results showed that the trajectory was close to the time optimal trajectory. Kelly et al.[33] and Rucco et al.[34] used sequential quadratic programming (SQP) and projection operator Newton method numerical optimization control methods to obtain the time optimal trajectory, respectively. This type of trajectory planning methods require a large number of complex

Automotive New Technology Research Institute, BYD Auto Industry Co., Shenzhen 518118, China. ✉email: zhang.dongmin@163.com

vehicle model parameters, and any change on parameters has a significant impact on the generated trajectory, resulting in a larger computational workload. The geometric characteristics of the trajectory also have an impact on lap time[4]. Braghin et al.[35] calculated the shortest path and minimum curvature path first and then combined them together with a single weight coefficient to generate the best raceline. Based on Braghin's work, Cardamone et al.[36] decomposed the track into several segments and used a group of weight coefficients to generate the best raceline. The deficiency of both works is that the obtained raceline is always not very smooth as the shortest path turns sharply at corners. Kapania et al.[37] decomposed trajectory planning into two steps to solve the problem of high computational complexity in trajectory planning algorithms: generating the speed curve with minimum time and generating the minimum curvature trajectory. The experimental results showed that this method can quickly generate the optimal trajectory compared with the nonlinear gradient descent algorithm[38]. Heilmeier et al.[39] proposed a method for generating minimum curvature trajectory using quadratic programming (QP), which can greatly reduce algorithm parameters and improve calculation speed, and found that minimum curvature trajectory is close to minimum time trajectory. Li et al. proposed Curvature-Integrated MPCC[40] which achieved a mean projected velocity of 93.18% of vehicle handling limits in real-world experiments, and Curvature-Integrated MPCC[41] which achieved a 11.4%-12.5% reduction in lap time compared to traditional MPCC.

This paper focuses on global minimum time trajectory planning for track racing based on offline high-precision map of track and refers to Heilmeier's work[39], considers curvature and distance factors in objective function, which is different from Braghin's[35] and Cardamone's[36] works, which just combine the shortest path and minimum curvature path directly. In order to make full use of the width of track, drivable curbs, size of vehicle and safe distance are taken into consideration carefully. Hyper petrol and electric racecars, and different driver models are simulated in CarSim to decide the best weight coefficient that balances the curvature and distance factors in minimum time trajectory. Simulation results show that the minimum time trajectories for different vehicles and drivers have significant difference.

Track mapping and post-processing

The methods of mapping can be divided into online and offline. For autonomous driving on ordinary roads, real-time and online perception of the surrounding environment and obtaining the boundary coordinates of the drivable area are of great significance. The existing methods can be summarized as using radar, cameras and other devices to detect objects and free space, and then post-processing to obtain boundary coordinates. This method can adapt to ever-changing environments and its accuracy can basically meet the needs of daily driving. However, this method also has some drawbacks: (1) It has high requirements for onboard sensors, computing power, and algorithms; (2) For safety reasons, uncertain areas are generally classified as non-drivable areas, so the perceived drivable area is generally smaller than the actual one. For driving on track, the boundary conditions of the track are almost unchanged, and track driving requires to fully utilize the width of the track. Therefore, online perception and mapping is unnecessary and inappropriate, while mapping offline is a better choice for track driving.

We conducted offline mapping of a circuit in China. Shown as Fig. 1a, b, LiDAR, panoramic cameras, RTK and other devices were used to obtain the high-precision three-dimensional (3D) coordinates of the track's boundaries, as well as the types and positions of curbs on both sides. Shown as Fig. 1c–f, there are 6 types of boundaries marked by different colors in the circuit, including normal road, paint, scraping curb, standard curb, sausage curb and carpet. Normal road and paint marked by green are smooth and non-slip which means they can be fully taken advantaged to optimize trajectory. Scraping curb and standard curb marked by yellow can be taken on without difficult if the suspension and tires can suffer the jolt. Sausage curb and carpet marked by red are dangerous and must be avoided. The distribution of different boundaries is shown as Fig. 2a, in which the green lines represent normal road and paint boundaries; the yellow lines represent scraping curb and standard curb; and the red lines represent sausage curb and carpet boundaries. For safety, it's necessary to set different safe distances for different boundaries depended on the driving skills. Here, safe distances 0.05 m, 0.05 m and 0.1 m are chosen for green, yellow and red boundaries. According to racing rules, at least two wheels should be on the inner side of the white solid lines (boundaries of racetrack). We consider the width of vehicle is 2 m, then the distance from the white solid lines to boundaries in Fig. 2a should be no more than 2 m. After post-processing, the boundaries of all the drivable area are shown as the black solid lines in Fig. 2b and the enlarged figure inside the blue square is shown in Fig. 2c.

Existing gridding method is using normals to the centreline of track, which is easy to adapt to different race environments and shows robust performance on various tracks. However, a disadvantage of this method is that the discretization step size between two points can not be too small (e.g. step size should be larger than 2.96 m in the circuit), otherwise some normals would be crossed at tight corner with a large track width[39] shown as Fig. 3a. Therefore, we proposed a new gridding method shown as Fig. 3b, which doesn't require normals to the centreline, but links points on inner boundary and outer boundary directly, the only requirement is that the distribution of red lines in Fig. 3b should be as even as possible. There is no limit to the discretization step size of our method, and the average step size is only 0.2 m in this work.

Trajectory planning

In this section, the trajectory node coordinates will be expressed by boundary coordinates and variables a in "Notation of raceline points"; then the quadratic expression for distance factor and curvature factor of the trajectory will be derived in "Description of distance factor" and "Description of curvature factor". In "Objective function for minimum time trajectory", the objective function for minimum time trajectory will be given, whose optimization variables are the position of the trajectory nodes (the variables a), and its constraint condition will

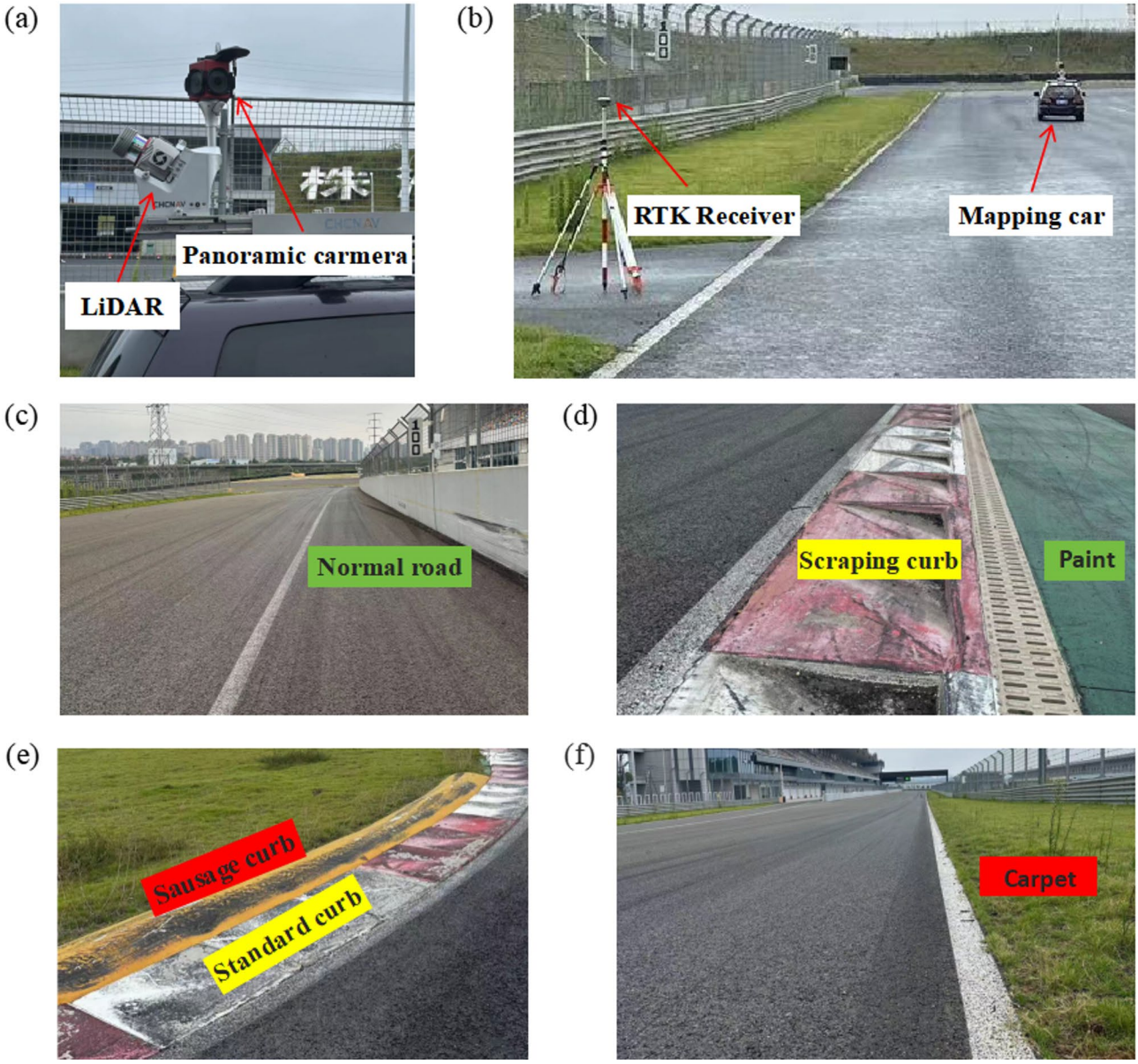

Fig. 1. Offline mapping method and different curbs. (**a**,**b**) mapping equipment; (**c**–**f**) represent different curbs, in which green means smooth and non-slip, yellow means safe but not comfortable, and red means dangerous and must be avoid.

be introduced in "Constraint condition". Finally, the iterative solution method of the optimal trajectory will be described in "Trajectory calculation and iteration".

Curvature and distance are the most important factors affecting the lap time following raceline. If the performance of vehicle is extremely weak, the best path for which is apt to shortest path; otherwise, if the performance is powerful enough, the vehicle should run along the minimum curvature path. Trajectory planning for cars in reality, both curvature and distance need to be considered to achieve a better lap time. QP method is a quick and robust method to calculate optimal trajectory. As long as the optimization problem is written as Eqs. (1.1) or (1.2), the Quadprog function integrated in MATLAB or "OSQP" solver[42] proposed by Stellato et al. can be used directly.

$$\min_{\mathbf{a}} J = \min_{\mathbf{a}} \left(\frac{1}{2}\mathbf{a}^{\mathrm{T}}\mathbf{H}\mathbf{a} + \mathbf{f}^{\mathrm{T}}\mathbf{a} \right) \ , \quad \text{subject to} \begin{cases} \mathbf{\Psi a} \leqslant \varphi \\ \mathbf{\Psi}_{eq}\mathbf{a} = \varphi_{eq} \\ \mathbf{lb} \leqslant \mathbf{a} \leqslant \mathbf{ub} \end{cases} , \tag{1.1}$$

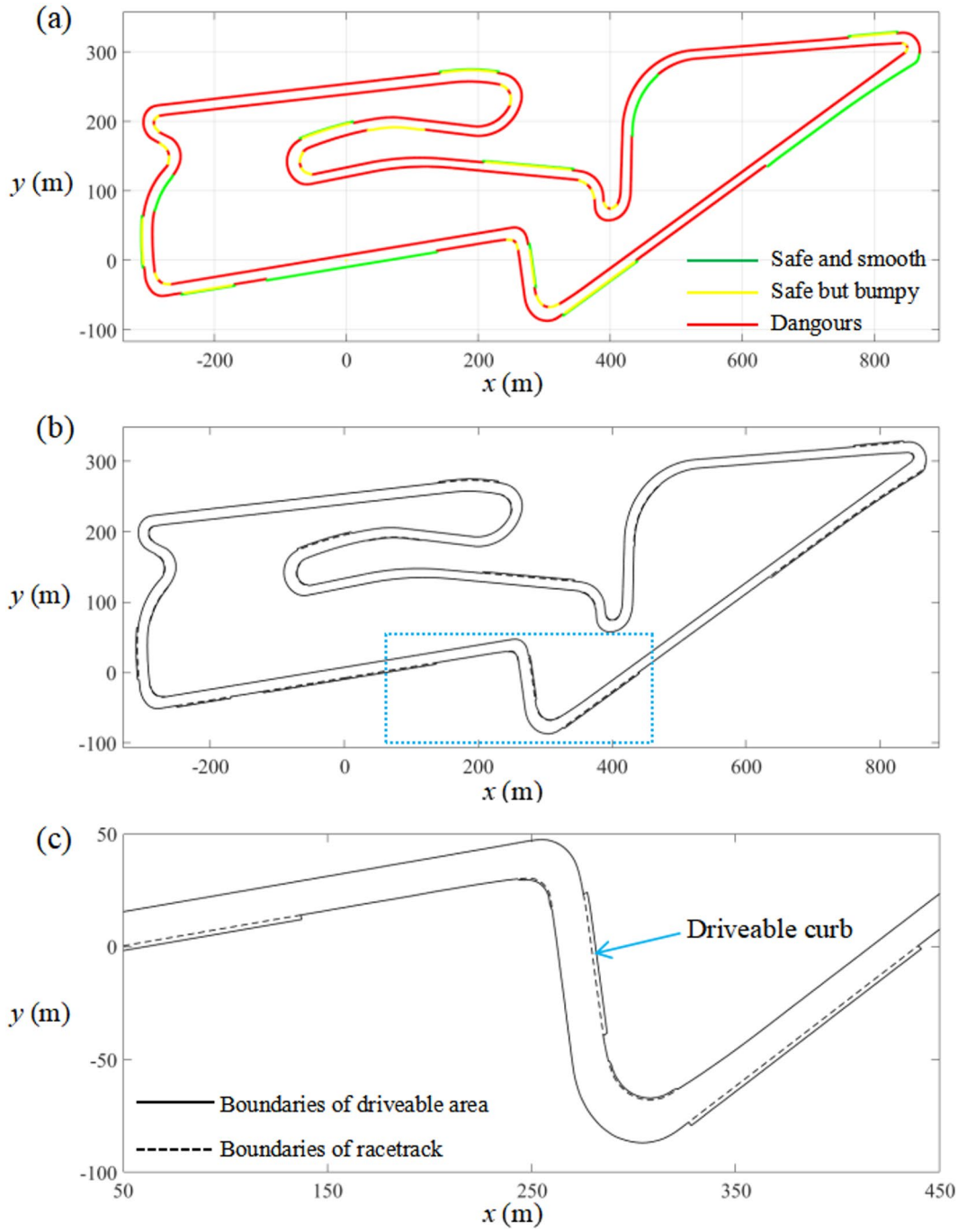

Fig. 2. Boundary post-processing considering racetrack and all drivable curbs. (**a**) Three kinds of curbs of the circuit; (**b**) drivable area including racetrack and drivable curbs with enough safe distance; (**c**) magnification view of drivable curbs.

$$\min_{\mathbf{a}} J = \min_{\mathbf{a}} \left(\frac{1}{2}\mathbf{a}^{\mathrm{T}}\mathbf{H}\mathbf{a} + \mathbf{f}^{\mathrm{T}}\mathbf{a} \right) , \quad \text{subject to } \mathbf{\Psi a} \leqslant \varphi , \tag{1.2}$$

where $\boldsymbol{a} \in \mathbf{R}^N$ is the decision variables.

Notation of raceline points

Shown as Fig. 4, the notation of the ith raceline point r_i can be expressed as

$$r_i = p_i + a_i v_i, \tag{2.1}$$

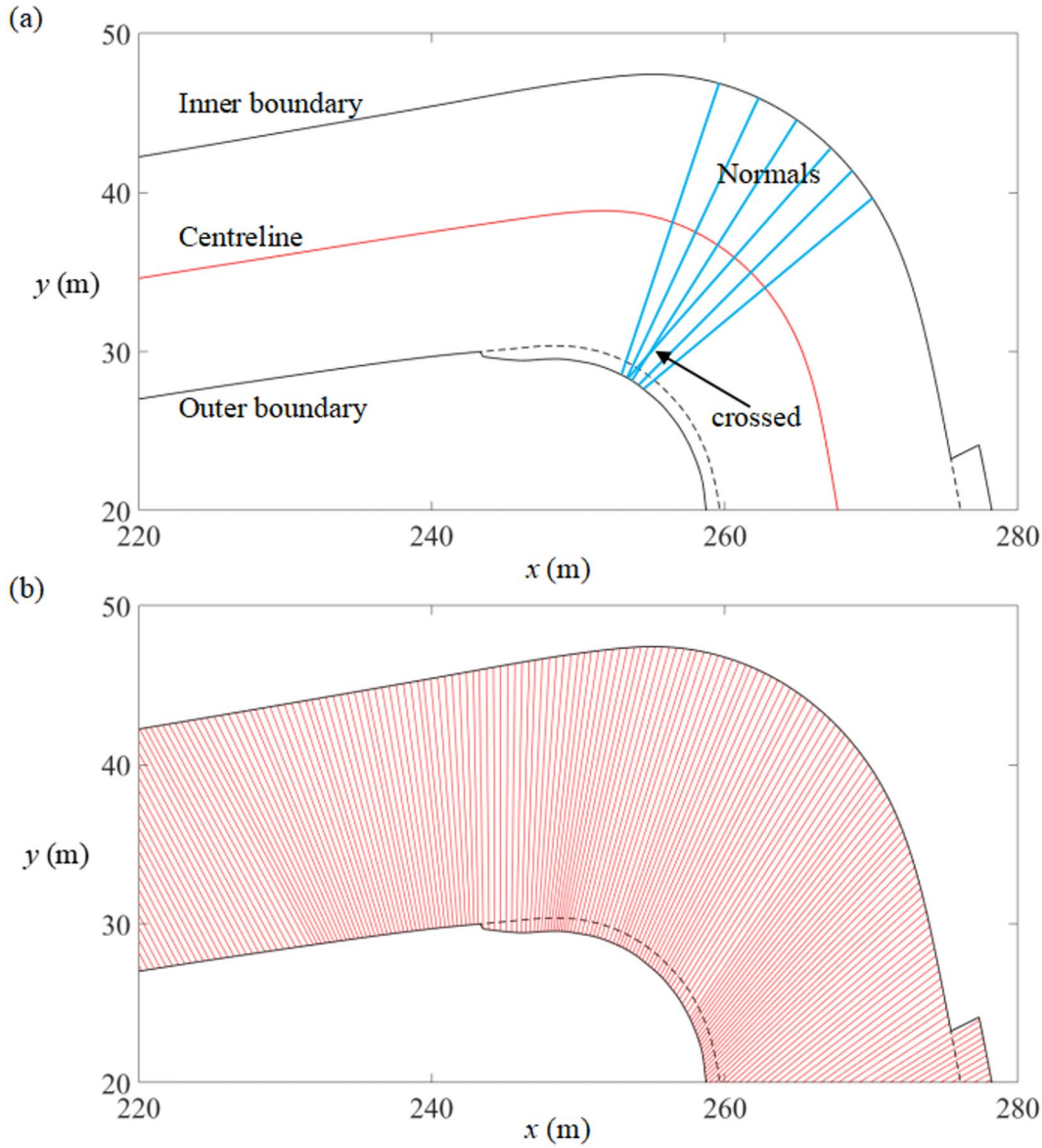

Fig. 3. Comparison of (**a**) existing gridding method and (**b**) our method. Normals would be crossed at tight corner with a large track width in existing gridding method; our method doesn't require normals to the centreline, but links points on inner boundary and outer boundary directly.

$$v_i = q_i - p_i, \tag{2.2}$$

$$\mathbf{r} = \mathbf{p} + \mathbf{va}, \tag{2.3}$$

where p_i and q_i are the inner boundary point and outer boundary point, and v_i is the vector from p_i to q_i. The shape of raceline is controlled by scalar a_i, and $a_i \in [0, 1]$ because the raceline must be inside the drivable area. Equation (2.1) can be written by matrix form as Eq. (2.3), in which

$$\mathbf{r} = [\ r_1 \ \cdots \ r_i \ \cdots \ r_N\ ]^{\mathrm{T}},\ \mathbf{p} = [\ p_1 \ \cdots \ p_i \ \cdots \ p_N\ ]^{\mathrm{T}},\ \mathbf{a} = [\ a_1 \ \cdots \ a_i \ \cdots \ a_N\ ]^{\mathrm{T}},$$
$$\mathbf{v} = \begin{bmatrix} q_1 - p_1 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ 0 & \cdots & q_i - p_i & \cdots & 0 \\ \vdots & \cdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & q_N - p_N \end{bmatrix}. \tag{3}$$

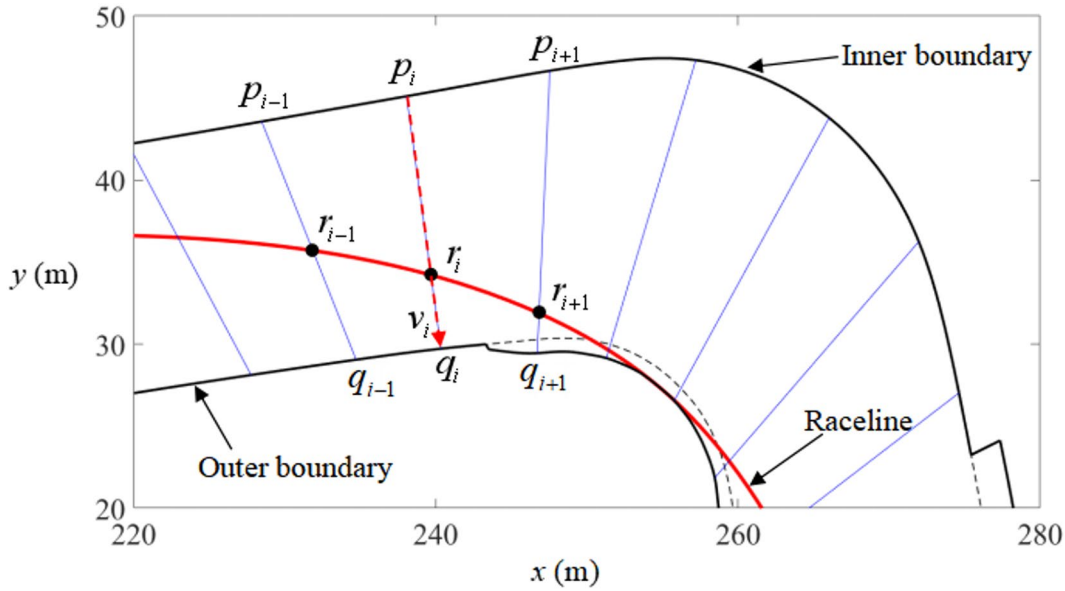

Fig. 4. Notation method of the raceline points by the points on inner boundary point and outer boundary.

Description of distance factor

In order to deal with the optimisation problem by QP method, we need to calculate the total discrete squared distance of raceline, for closed loop that is

$$J_s = \sum_{i=1}^{N} {s_i}^2 = \sum_{i=1}^{N-1} (r_{i+1} - r_i)^2 + (r_1 - r_N)^2, \tag{4}$$

where s_i can be written in matrix form,

$$\mathbf{s} = \begin{bmatrix} -1 & 1 & \cdots & 0 & 0 \\ 0 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 1 \\ 1 & 0 & \cdots & 0 & -1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \\ \cdots \\ r_{N-1} \\ r_N \end{bmatrix} = \mathbf{Ar}. \tag{5}$$

Thus the matrix form for Eq. (4) is

$$J_s = \mathbf{s}^{\mathrm{T}}\mathbf{s} = \frac{1}{2}\mathbf{a}^{\mathrm{T}}\mathbf{H}_s\mathbf{a} + {\mathbf{f}_s}^{\mathbf{T}}\mathbf{a} + c_s, \tag{6}$$

in which c_s is a constant and can be ignored in QP problem, and

$$\mathbf{H}_s = 2\mathbf{v}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{Av},\ \ \mathbf{f}_s = 2\mathbf{v}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{Ap}. \tag{7}$$

Inserting the x and y coordinates of inner boundary points $\mathbf{p}$ and outer boundary points $\mathbf{q}$ into Eq. (7) leads to the objective function of 2D minimum distance trajectory, which is

$$J_s = \frac{1}{2}\mathbf{a}^{\mathrm{T}}\left(\mathbf{H}_{s,x} + \mathbf{H}_{s,y}\right)\mathbf{a} + \left(\mathbf{f}_{s,x} + \mathbf{f}_{s,y}\right)^{\mathbf{T}}\mathbf{a} + c_{s1}. \tag{8}$$

Description of curvature factor

Similar to distance factor, we also need to minimize the discrete squared curvature of raceline, which is

$$J_\kappa = \sum_{i=1}^{N} {\kappa_i}^2 = \sum_{i=1}^{N} \frac{\left({x_i}'{y_i}'' - {y_i}'{x_i}''\right)^2}{\left({x_i}'^2 + {y_i}'^2\right)^3}, \tag{9}$$

in which x_i and y_i are the x and y coordinates of r_i, and κ_i^2 can be rewritten as

$$\kappa_i{}^2 = \frac{y_i{}'^2}{\left(x_i{}'^2 + y_i{}'^2\right)^3} x_i{}''^2 - \frac{2x_i{}'y_i{}'}{\left(x_i{}'^2 + y_i{}'^2\right)^3} x_i{}'' y_i{}'' + \frac{x_i{}'^2}{\left(x_i{}'^2 + y_i{}'^2\right)^3} y_i{}''^2 = T_{xx,i} x_i{}''^2 + T_{xy,i} x_i{}'' y_i{}'' + T_{yy,i} y_i{}''^2. \quad (10)$$

Thus J_κ can be rewritten in matrix form as

$$J_\kappa = \left(\mathbf{x}''\right)^{\mathrm{T}} \mathbf{T}_{xx} \mathbf{x}'' + \left(\mathbf{x}''\right)^{\mathrm{T}} \mathbf{T}_{xy} \mathbf{y}'' + \left(\mathbf{y}''\right)^{\mathrm{T}} \mathbf{T}_{yy} \mathbf{y}'', \quad (11)$$

where $\mathbf{x}'' = [x_1{}'' \ldots x_i{}'' \ldots x_N{}'']^{\mathrm{T}}$ and $\mathbf{y}'' = [y_1{}'' \ldots y_i{}'' \ldots y_N{}'']^{\mathrm{T}}$ are the vector notations for the second derivative of the coordinates at each discretization point, and $\mathbf{T}_{xx}$, $\mathbf{T}_{xy}$ and $\mathbf{T}_{yy}$ are diagonal matrices only consisting of the first derivative x' and y', which are

$$\mathbf{T}_{xx} = \begin{bmatrix} \frac{y_1{}'^2}{\left(x_1{}'^2+y_1{}'^2\right)^3} & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ 0 & \cdots & \frac{y_i{}'^2}{\left(x_i{}'^2+y_i{}'^2\right)^3} & \cdots & 0 \\ \vdots & \cdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \frac{y_N{}'^2}{\left(x_N{}'^2+y_N{}'^2\right)^3} \end{bmatrix}, \quad (12.1)$$

$$\mathbf{T}_{xy} = \begin{bmatrix} \frac{-2x_1{}'y_1{}'}{\left(x_1{}'^2+y_1{}'^2\right)^3} & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ 0 & \cdots & \frac{-2x_i{}'y_i{}'}{\left(x_i{}'^2+y_i{}'^2\right)^3} & \cdots & 0 \\ \vdots & \cdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \frac{-2x_N{}'y_N{}'}{\left(x_N{}'^2+y_N{}'^2\right)^3} \end{bmatrix}, \quad (12.2)$$

$$\mathbf{T}_{yy} = \begin{bmatrix} \frac{x_1{}'^2}{\left(x_1{}'^2+y_1{}'^2\right)^3} & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \cdots & \vdots \\ 0 & \cdots & \frac{x_i{}'^2}{\left(x_i{}'^2+y_i{}'^2\right)^3} & \cdots & 0 \\ \vdots & \cdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \frac{x_N{}'^2}{\left(x_N{}'^2+y_N{}'^2\right)^3} \end{bmatrix}. \quad (12.3)$$

According to Alexander et al.[39], the matrices $\mathbf{T}_{xx}$, $\mathbf{T}_{xy}$ and $\mathbf{T}_{yy}$ can be assumed to be constants as x' and y' are approximately constant since path heading only changes slightly compared to the reference line as long as it is tightly constrained by the inner and outer track boundaries. Therefore, at the first time, we can obtain x' and y' from inner boundary of racetrack corresponding to the reference line, and in the next iteration, the trajectory calculated in the previous iteration is used as the reference line to calculate x' and y'.

The next question is how to express the second derivatives $\mathbf{r}'$ (including $\mathbf{x}''$ and $\mathbf{y}''$) in terms of the decision variables $\mathbf{a}$. The coordinates of nodes on trajectory can be fitted by a third order spline, which is

$$l_i(t) = \lambda_{0,i} + \lambda_{1,i} t + \lambda_{2,i} t^2 + \lambda_{3,i} t^3, \ t = [0, \ 1], \quad (13)$$

where $\lambda_{0,i}$, $\lambda_{1,i}$, $\lambda_{2,i}$ and $\lambda_{3,i}$ are coefficients need to be solved, and point on trajectory r_i is the control point of the splines. According to the continuity of spline, we have

$$l_i(0) = r_i, \ l_i(1) = r_{i+1}, \ l_i{}'(1) = l_{i+1}{}'(0), \ l_i{}''(1) = l_{i+1}{}''(0). \quad (14)$$

Inserting Eq. (13) into Eq. (14) yields

$$\begin{aligned} &\lambda_{0,i} = r_i, \ \lambda_{0,i} + \lambda_{1,i} + \lambda_{2,i} + \lambda_{3,i} = r_{i+1}, \\ &\lambda_{1,i} + 2\lambda_{2,i} + 3\lambda_{3,i} = \lambda_{1,i+1}, \ 2\lambda_{2,i} + 3\lambda_{3,i} = \lambda_{2,i+1}. \end{aligned} \quad (15)$$

Let m_i to represent the second derivative of l_i at $t = 0$, thus $m_i = 2\lambda_{2,i}$, then inserting m_i into Eq. (15) leads to the solutions of $\lambda_{0,i}$ - $\lambda_{4,i}$ and the relation between r_i and m_i,

$$\lambda_{0,i} = r_i, \ \lambda_{1,i} = r_{i+1} - r_i - \frac{m_i}{3} - \frac{m_{i+1}}{6}, \ \lambda_{2,i} = \frac{m_i}{2}, \ \lambda_{3,i} = \frac{m_{i+1} - m_i}{6} \quad (16.1)$$

$$r_{i+1} - 2r_i + r_{i-1} = \frac{m_{i+1} + 4m_i + m_{i-1}}{6}, \ i = [2, \ N-1]. \quad (16.2)$$

For closed loop, the relation between r_i and m_i when $i=1$ and N are

$$r_2 - 2r_1 + r_N = \frac{m_2 + 4m_1 + m_N}{6},\ i = 1, \tag{17.1}$$

$$r_1 - 2r_N + r_{N-1} = \frac{m_1 + 4m_N + m_{N-1}}{6},\ i = N. \tag{17.2}$$

Combining Eqs. (16.2) and (17) leads to the second derivatives $\mathbf{m}$, which is expressed as

$$\mathbf{m} = \mathbf{B}^{-1}\mathbf{C}\mathbf{r} = \mathbf{B}^{-1}\mathbf{C}\left(\mathbf{p} + \mathbf{v}\mathbf{a}\right), \tag{18}$$

in which

$$\mathbf{B} = \begin{bmatrix} 4 & 1 & 0 & \cdots & 0 & 0 & 1 \\ 1 & 4 & 1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 4 & 1 \\ 1 & 0 & 0 & \cdots & 0 & 1 & 4 \end{bmatrix}, \mathbf{C} = \begin{bmatrix} -12 & 6 & 0 & \cdots & 0 & 0 & 6 \\ 6 & -12 & 6 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 6 & -12 & 6 \\ 6 & 0 & 0 & \cdots & 0 & 6 & -12 \end{bmatrix}. \tag{19}$$

Inserting Eq. (18) into Eq. (11) gives,

$$J_\kappa = \frac{1}{2}\mathbf{a}^{\mathrm{T}}\mathbf{H}_\kappa\mathbf{a} + \mathbf{f}_\kappa{}^{\mathrm{T}}\mathbf{a} + c_\kappa, \tag{20}$$

in which,

$$\mathbf{H}_\kappa = 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_x\right)^{\mathrm{T}}\mathbf{T}_{xx}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_x\right) + 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_x\right)^{\mathrm{T}}\mathbf{T}_{xy}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_y\right) + 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_y\right)^{\mathrm{T}}\mathbf{T}_{yy}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{v}_y\right), \tag{21.1}$$

$$\mathbf{f}_\kappa = 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_x\right)^{\mathrm{T}}\mathbf{T}_{xx}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_x\right) + 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_x\right)^{\mathrm{T}}\mathbf{T}_{xy}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_y\right) + 2\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_y\right)^{\mathrm{T}}\mathbf{T}_{yy}\left(\mathbf{B}^{-1}\mathbf{C}\mathbf{p}_y\right). \tag{21.2}$$

Objective function for minimum time trajectory

Both curvature and distance can affect the lap time following trajectory. The weight coefficient between both factors should vary with vehicle's properties and driving skills. In this paper, the objective function for minimum time trajectory is set as

$$J = J_\kappa + \varepsilon J_s = \frac{1}{2}\mathbf{a}^{\mathrm{T}}\left(\mathbf{H}_\kappa + \varepsilon\mathbf{H}_s\right)\mathbf{a} + \left(\mathbf{f}_\kappa + \varepsilon\mathbf{f}_s\right)^{\mathrm{T}}\mathbf{a} + c, \tag{22}$$

where $\varepsilon \geq 0$ is the weight coefficient, which can be a constant scalar or a diagonal matrix with varying elements at different nodes, in this case, the ratio of curvature and distance factors can be tuned point by point, which gives the possibility of local optimization. But in this work, we only discuss a constant weight coefficient ε.

Constraint condition

Vehicle should run inside the drivable area including racetrack and drivable curbs. When planning the trajectory of vehicle's centre point, vehicle size must be considered and a series of safe distances are better to be considered especially near to dangerous curbs. Shown as Fig. 5, in the width direction of racetrack, the furthest distance of vehicle body to the center point is

$$w_{\mathrm{v}} = \frac{\sqrt{{l_v}^2 + {b_v}^2}}{2}\cos\left(\frac{v_i \cdot d_i}{|v_i|\,|d_i|} - \arctan\frac{b_v}{l_v}\right) \tag{23}$$

in which the symbol "| |" means L2 norm of a vector, l_v and b_v are the length and width of vehicle, d_i is the vector from point r_i to r_{i+1} (heading direction of vehicle), which can be replaced appropriately by the ith and $(i+1)^{\mathrm{th}}$ points on reference line. The distance form trajectory to boundaries should be not less than the sum of w_v and w_s, thus

$$\frac{\left(w_{\mathrm{v}} + w_{\mathrm{s},i}\right)|v_i|}{n_{Ii} \cdot v_i} \leqslant a_i \leqslant 1 - \frac{\left(w_{\mathrm{v}} + w_{\mathrm{s},i}\right)|v_i|}{n_{Oi} \cdot v_i}, \tag{24}$$

where $w_{s,i}$ is the safe distance at the ith node, n_{Ii} and n_{Oi} are unit length normal vectors of inner boundary at p_i and outer boundary at q_i, respectively.

Trajectory calculation and iteration

The objective function and constraint condition for minimum time trajectory have been derived. Set a value for weight coefficient ε, then the "quadprog" function integrated in MATLAB can be used to calculate the decision variables $\mathbf{a}$, which can give a series of discrete points on trajectory. As mentioned in Sect. 3.3, $\mathbf{T}_{xx}$, $\mathbf{T}_{xy}$ and $\mathbf{T}_{yy}$ are

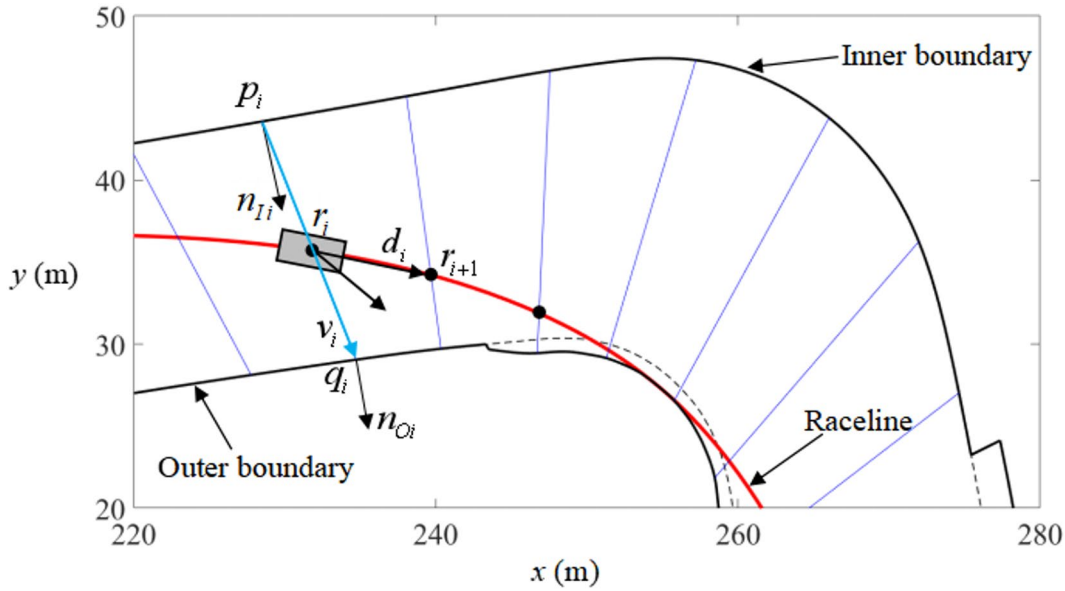

Fig. 5. Constraint condition on the decision variables **a** considering the size of vehicle and safe distance.

assumed to be constants and calculated based on a reference line. A closer reference line to the optimal trajectory can make the assumption more accurate. In the first calculation, the inner boundary of racetrack is used as the reference line, then the calculated trajectory will be the reference line for the next iteration. After a few iterations, the trajectory will remain basically unchanged, and the last iteration will produce the optimal trajectory under the selected weight coefficient ε.

The iterated trajectories when $\varepsilon = 0$ are shown in Fig. 6a, in which the green line represents the first iterated trajectory which is not very smooth and far from the final optimal trajectory, but trajectories will get stable after several iterations, which can be seen clearly from Fig. 6b that trajectories of 3rd to 6th iterations are almost totally coincide. The sum of distance and curvature factors are listed in Table 1, as iteration increases, the sum of distance factor (s_i or s_i^2) changes slightly, while the sum of curvature factor ($|\kappa_i|$ or κ_i^2) decreases quickly at first and then gets stable after 4 iterations, therefore, iterating 4 times is effective and enough to obtain a rewarding optimal trajectory.

Figure 7a, b show the influence of weight coefficient ε on optimal trajectory iterated 4 times. When $\varepsilon = 0$, the trajectory is equal to minimum curvature path, which takes fully advantage of the width of track but goes too far at some places, and when ε tends to infinity, the trajectory tends to minimum distance path, which prefers to direct lines but goes too sharp at corners. Our method when $\varepsilon = 0$ and Braghin's method[35] when its weight coefficient equals 0.15 are compared in Fig. 7c, which can be observed that both trajectories are coincide at most place but the trajectory by Braghin's method goes too far when $y \in [230, 250]$. More details can be seen in Table 2, as weight coefficient increases, the sum of distance factor decreases while the sum of curvature factor increases. Comparing both methods can find that the sum of distance and the sum of curvature of our method are both better than Braghin's method.

Simulation and discussion

Minimum time trajectory is depended on vehicle's performance and driver's level. In this section, in order to find the best weight coefficient ε for different vehicle and driver, all tests are made both on the original vehicle in CarSim and a conversion vehicle based on the original vehicle, and in every test, the global trajectory or the driver model is changed. The global trajectory is controlled by weight coefficient ε and the driver model is selected in CarSim but four acceleration limits are modified in different test.

The settings of all simulation tests considered in this work are listed in Table 3. In order to compare conveniently, we selected a hyper petrol racecar (Exotic Racecar - MT, Var. Gains, Pedal Force) in CarSim, then we changed its sprung mass into 2200 kg and replaced a 960 kw AWD powertrain system to simply build a hyper electric racecar. "No Offset, 0.5 s Preview" was selected as the driver path follower, and the target speed was calculated based on a modified "Race Driving" listed in Table 3, whose maximum acceleration limits at four directions vary from 0.8 to 1.8 g, which can reflect the performance of vehicle and the level of driving.

At first, drivable curbs are considered in our trajectory planning method, and the simulated lap time comparisons between taking advantage of drivable curbs and only driving on racetrack are shown as Fig. 8a, from which we can conclude that the lap time of trajectory considering drivable curbs is much faster than trajectory only using racetrack, and as the acceleration increases, the lap time decreases. From Fig. 8b, the maximum lap time improvements for petrol and electric vehicle in these tests are 0.84 s and 0.62 s, and the average improvements are 0.68 s and 0.46 s, respectively. As acceleration limits increase, the gains from taking advantage of drivable curbs also increase.

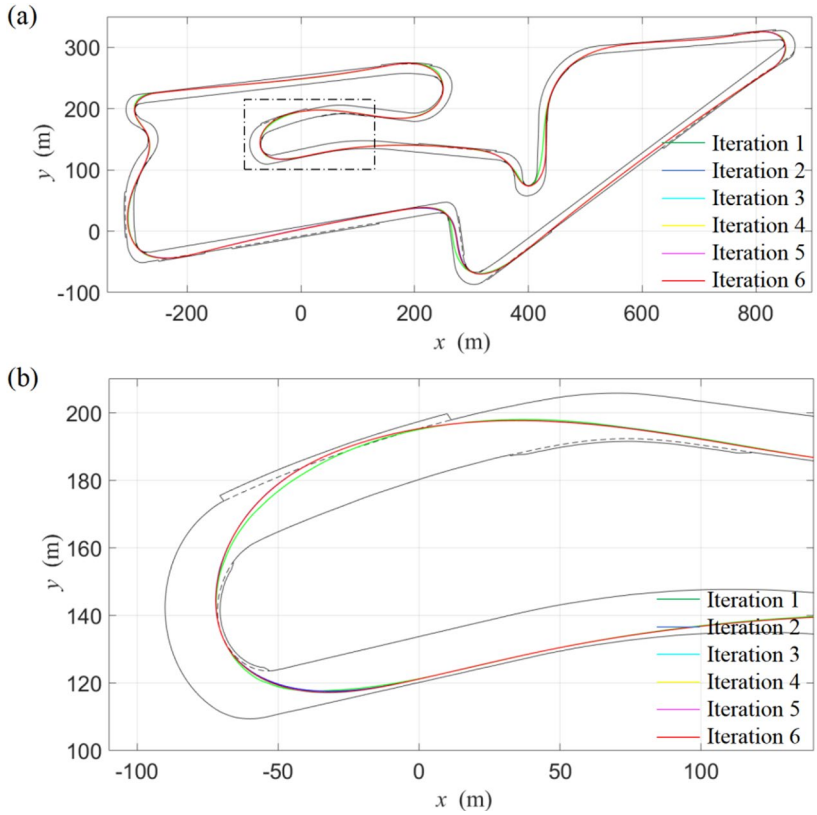

Fig. 6. Evolution process of minimum time trajectory as iteration increases, and iterating 4 times is effective and enough to obtain a rewarding optimal trajectory. (**a**) Global 2D view of trajectories; (**b**) local magnification view of trajectories around a bend.

Iterations	1	2	3	4	5	6
Sum of $\lvert\kappa_i\rvert$ (m^{-1})	37.27	25.56	23.42	23.16	23.12	23.11
Sum of κ_i^2 (m^{-2})	3.117	1.267	0.761	0.641	0.626	0.608
Sum of s_i (m)	3668.69	3665.96	3668.35	3668.08	3668.06	3668.07
Sum of s_i^2 (m^2)	4540.91	4536.48	4541.96	4541.39	4541.56	4541.58

Table 1. Evolution process of distance and curvature factors as iteration increases.

Trajectories calculated by 11 different ε listed as group 2 in Table 3 were simulated. For each vehicle model and driver model, the lap time will be a little change depended on different ε, and when the time is shortest, the ε is the best and accordingly the path is the minimum time trajectory for the given vehicle and driver model. The simulated lap times are shown as Fig. 9a, from which we can observe that it's obviously the higher the acceleration limits, the better the lap time. Figure 9b, c displays the relationship between the best weight coefficient and acceleration limits, which is different for petrol racecar and electric racecar. For the given petrol racecar, if acceleration limits are larger than 1.0 g, the best weight coefficient $\varepsilon = 0$, which means the minimum curvature trajectory equals the minimum time trajectory; if acceleration limits are within 0.8–1.0 g, the minimum time trajectory is about 0.1 s faster than minimum curvature trajectory when $\varepsilon = 0.0002$. For the given electric racecar, when acceleration limits are smaller than 1.7 g, the best weight coefficient will increase as acceleration limits reduce, which means the weaker the vehicle or driver, the more the distance should be considered for optimal trajectory, and the minimum time trajectory can be 0.22 s faster than minimum curvature trajectory.

In order to explore the deep reasons that influence the lap time of petrol and electric racecars, we arranged the third group of simulation tests listed in Table 3, in which the acceleration limits at four directions (longitudinal: throttle Ax+, braking Ax-, and lateral: turning left and right Ay) are different. The simulation results are shown in Fig. 10, in which Fig. 10a–d display the settings of acceleration limits, the best lap time, the best weight coefficient and the lap time differences produced by weight coefficient under, respectively. From Fig. 10a, b we can observe that lateral acceleration limit Ay has the most remarkable influence on lap time while the influence of longitudinal acceleration limit Ax on lap time is not such obvious. If increase Ay from 1.0 g to 1.7 g, the average improvement on lap time is 12.56 s, but the same change on Ax + and Ax- can only bring 1.88 s and 2.39 s improvements into lap time, respectively. Figure 10a, c,d show that when Ay is huge enough (e.g. 1.7 g),

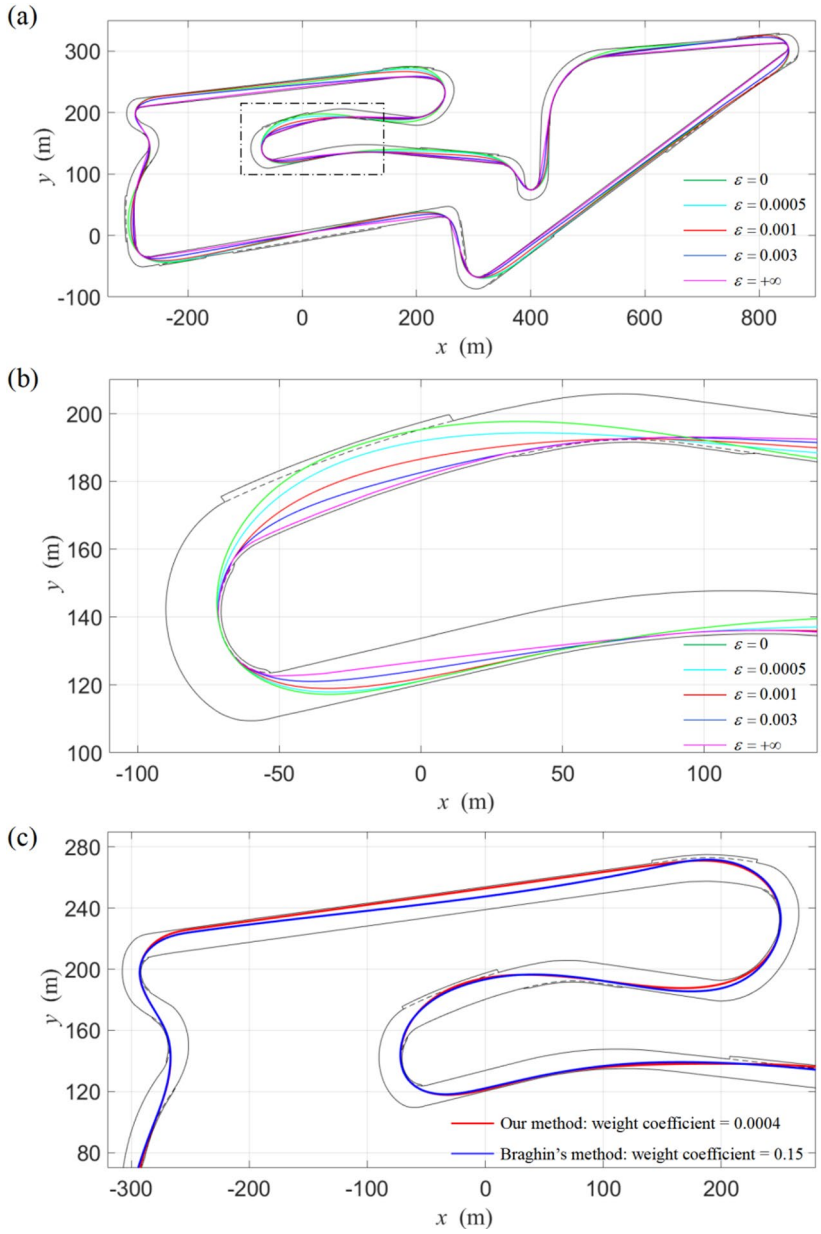

Fig. 7. Evolution process of minimum time trajectory as weight coefficient increases iterated 4 times. (**a**) global 2D view of trajectories; (**b**) local magnification view of trajectories around a bend; (**c**) comparison of our method when $\varepsilon = 0$ and Braghin's method[35] when its weight coefficient equals 0.15.

Method	Our method					Braghin's method[35]			
Weight coefficient	0	0.0005	0.001	0.003	+ ∞	0.05	0.10	0.15	0.20
Sum of $\lvert\kappa_i\rvert$ (m^{-1})	23.16	23.10	23.19	23.24	78.93	24.27	25.90	27.74	29.79
Sum of κ_i^2 (m^{-2})	0.641	0.737	0.769	0.885	8.182	0.702	0.779	0.902	1.072
Sum of s_i (m)	3668	3650	3635	3613	3595	3663	3658	3653	3648
Sum of s_i^2 (m^2)	4541	4495	4459	4402	4357	4527	4514	4501	4488

Table 2. Evolution process of distance and curvature factors as weight coefficient increases. The sum of distance and the sum of curvature of our method are both better than Braghin's method.

Simulation tests	Vehicle models: Exotic racecar—MT, Var. Gains, pedal force	Driver models: No offset, 0.5 s preview Modified race driving	Target paths: (Weight coefficient ε 10^{-4})
Group 1	Original vehicle (petrol, 300 kw RWD, Sprung mass 1200 kg)	Four acceleration limits are the same: 0.8 / 0.9 / 1.0 / 1.1 / 1.2 / 1.3 / 1.4 / 41.5 / 1.6 / 1.7 / 1.8 g	Drivable curbs YES: 0
Group 2			Drivable curbs NO: 0
Group 3	Modified vehicle (electric, 960 kw AWD, Sprung mass 2200 kg)	Acceleration limits of throttle (Ax+) / braking (Ax-)/turning (Ay): ① 1.7 / 1.7 / 1.7 g ② 1.0 / 1.7 / 1.7 g ③ 1.7 / 1.0 / 1.7 g ④ 1.0 / 1.0 / 1.7 g ⑤ 1.7 / 1.7 / 1.0 g ⑥ 1.0 / 1.7 / 1.0 g ⑦ 1.7 / 1.0 / 1.0 g ⑧ 1.0 / 1.0 / 1.0 g	Drivable curbs YES: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Table 3. Settings of all simulation tests considered in this work: the first group tests are used to compare the influence of drivable curbs; the second group tests are used to explore the influence of weight coefficient; the third group tests are used to analysis the influence of four acceleration limits.

distance factor doesn't need to be considered and the minimum time trajectory equals the minimum curvature trajectory, while if Ay is small enough (e.g. 1.0 g), considering distance factor could make the optimal trajectory faster, and in this case, the higher the longitudinal acceleration, the more lap time can be saved. When Ax = 1.7 g and Ay = 1.0 g, the minimum time trajectory is 0.42 s and 0.92 s faster than the minimum curvature trajectory for petrol and electric racecars.

We selected the 1st and 5th tests when $\varepsilon = 0$ and 0.0009 to analysis the reason why $\varepsilon = 0.0009$ is faster than 0 when Ay = 1.0 g but slower when Ay = 1.7 g. As shown in Fig. 11a, the peak curvature at apex of $\varepsilon = 0.0009$ is higher than $\varepsilon = 0$, but before and after the apex, the curvature of $\varepsilon = 0.0009$ is lower than $\varepsilon = 0$. This feature of $\varepsilon = 0.0009$ leads to a lower speed at apex, but a later breaking point before corner and an earlier accelerating point after apex if the break and powertrain are powerful enough. Although Ax limits are both 1.7 g in Fig. 11b, c, the tires cannot afford enough friction force to longitude when Ay = 1.7 g in Fig. 11b, while when Ay = 1.0 g, the tires can support powerful longitudinal acceleration, which allows later breaking and earlier acceleration in Fig. 11c. Therefore, the trajectory considering distance factor is faster when lateral acceleration limits is not very high and the tires can support additional powerful longitudinal acceleration. The critical value of lateral acceleration limits for different vehicle is different, which can be referred to Fig. 9.

Conclusions

A global minimum time trajectory planning method for racing based on offline high-precision map of track was proposed, in which the curvature and distance factors were both considered in the objective function. We firstly introduced offline mapping of track, post processing of drivable curbs and a new gridding method which doesn't require normal to the centerline and the discretization step size can be very small. In order to make full use of the width of track and ensure safety, drivable curbs, size of vehicle and different safe distances were taken into consideration carefully in constraint conditions. The sum of discrete squared distance and the sum of discrete squared curvature were derived and combined together by a weight coefficient to constitute the objective function, then the "quadprog" function embedded in MATLAB was used to calculate and iterate the target trajectories. A hyper petrol racecar and a simply built electric racecar, driver model with different acceleration limits and different trajectories were simulated by CarSim. Theoretic and simulated results show that:

1. Iteration makes trajectory smoother and more reasonable, and iterating 4 times is effective and enough to obtain a rewarding optimal trajectory;

2. If drivable curbs are used, the maximum/average improvements on lap time are 0.84/0.68 s and 0.62/0.46 s for the given petrol and electric racecar;

3. Minimum time trajectories for different vehicle and driver are different:

3.1 Considering distance factor is faster if acceleration limits are smaller than a critical value, which is totally different for the given petrol racecar (1.1 g) and electric racecar (1.7 g), otherwise, minimum time trajectory equals minimum curvature trajectory;

3.2 If longitude acceleration limits are much bigger than lateral acceleration limits, considering distance factor will be much faster, especially for electric racecar, for which at most 0.92 s can be saved.

The best weight coefficient that balances the curvature and distance factors in minimum time trajectory is depended on vehicle's performance and driver's level. Here we only discussed a constant weight coefficient, but it also allows different weight coefficient at different place, which gives the possibility to local optimization according to vehicle's feature and driver's habit.

To enable broader applications of optimal trajectory planning, more research efforts can be invested in enhancing real-time performance, improving robustness and safety, developing advanced optimization formulations, achieving multi-vehicle coordination, and enhancing capabilities for specific applications.

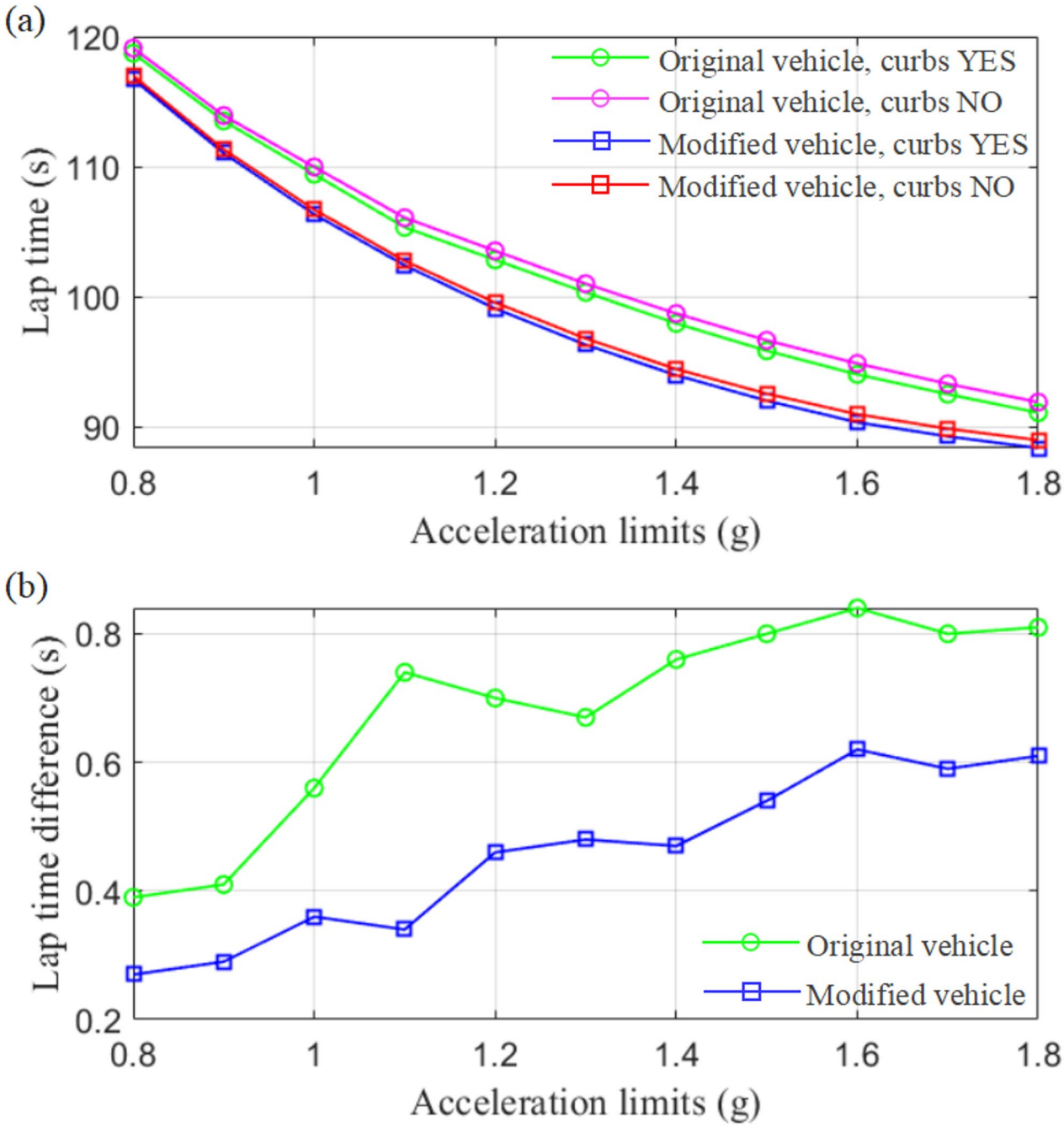

Fig. 8. (**a**) Lap time and (**b**) lap time difference comparisons between trajectories taking advantage of drivable curbs and only driving on racetrack. The maximum/average lap time improvements are 0.84/0.68 s and 0.62/0.46 s for petrol and electric vehicle if using drivable curbs.

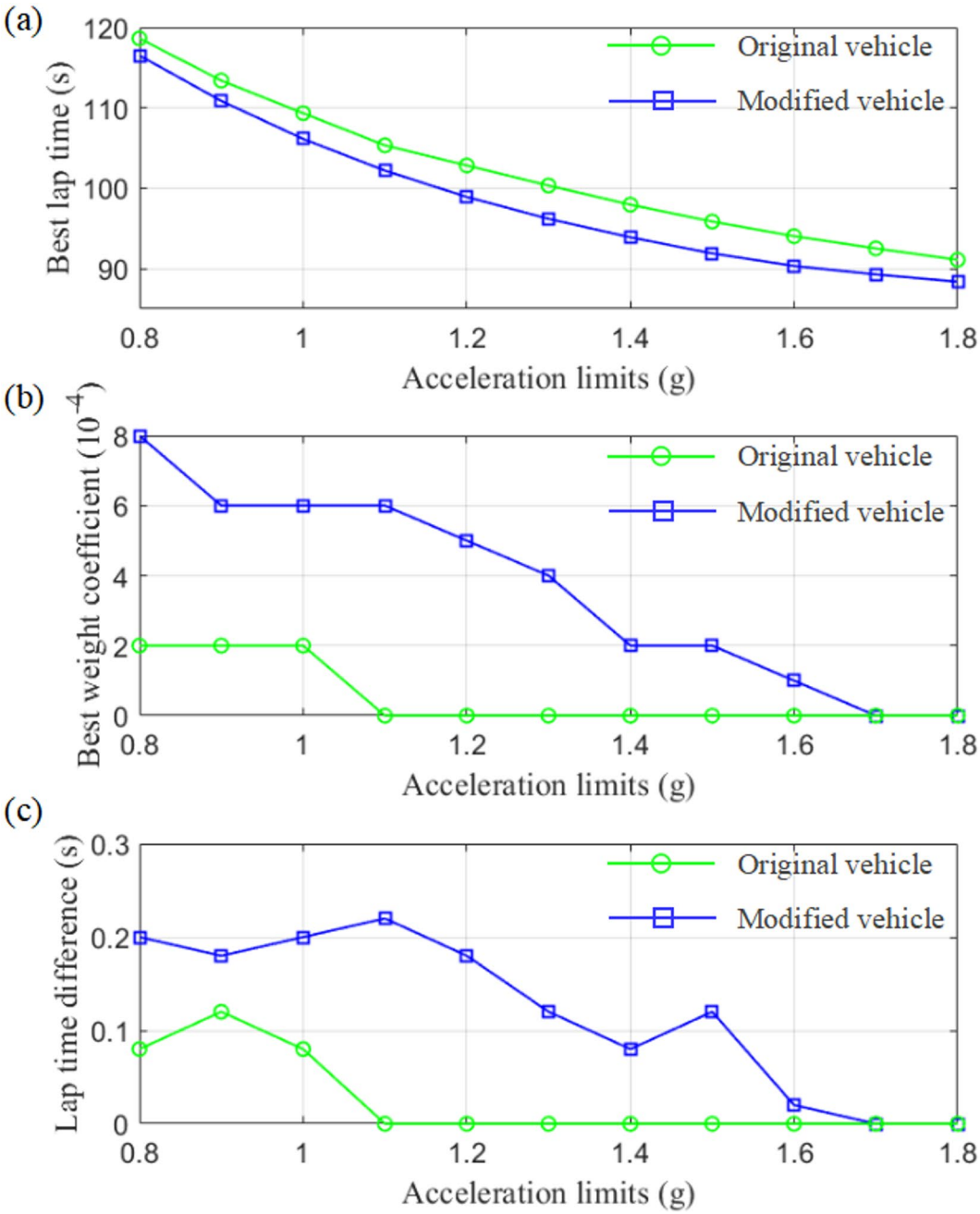

Fig. 9. (**a**) Lap time; (**b**) best weight coefficient and (**c**) lap time difference comparisons between original (petrol) and modified (electric) vehicles under different acceleration limits. Considering distance can bring 0.1 s and 0.22 s improvement into lap time for the given petrol and electric vehicle.

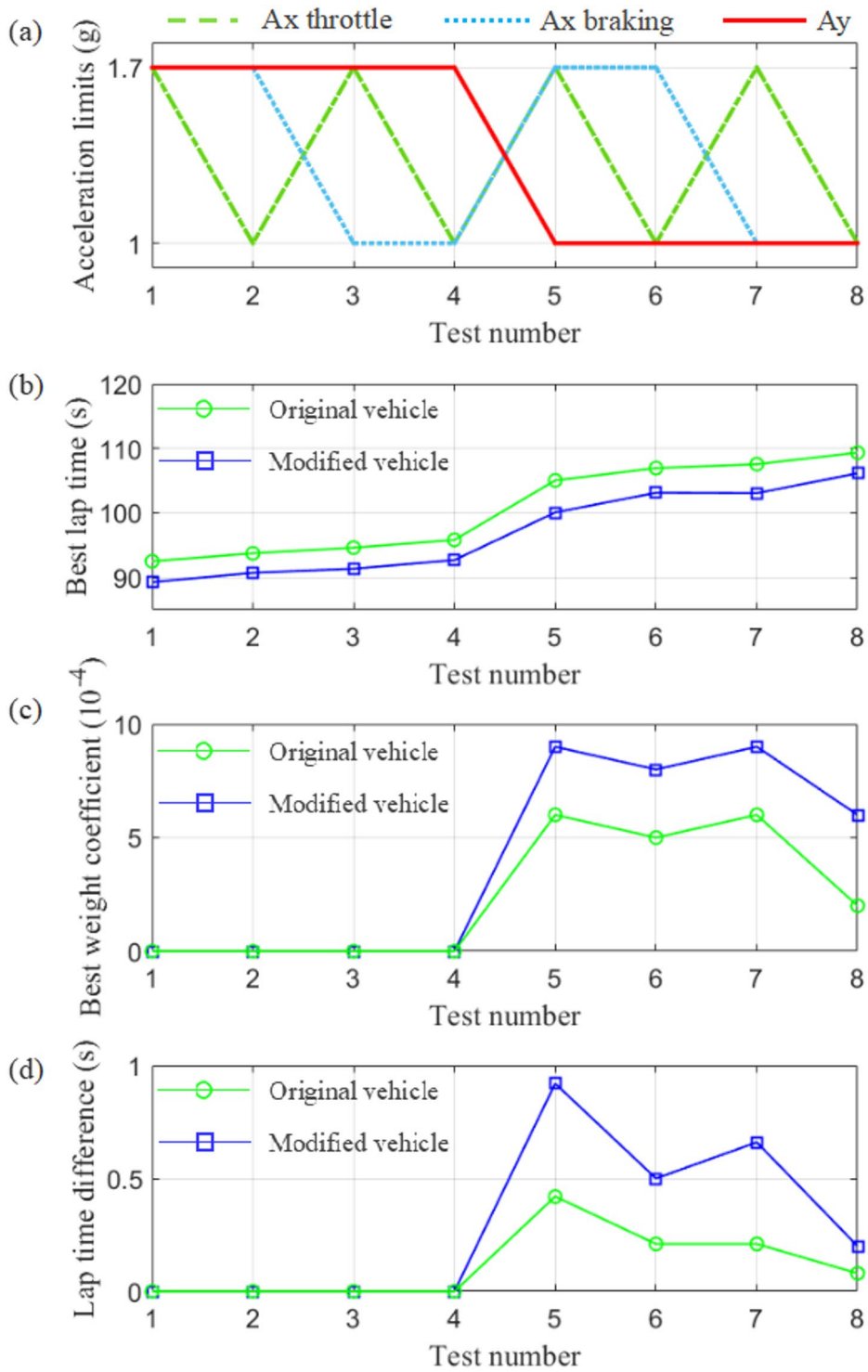

Fig. 10. (**a**) Settings of four acceleration limits; (**b**) lap time; (**c**) best weight coefficient and (**d**) lap time difference comparisons between original (petrol) and modified (electric) vehicles in different tests. Lateral acceleration limit Ay has remarkable influence on lap time, while the influence of longitudinal acceleration limit Ax is not such obvious.

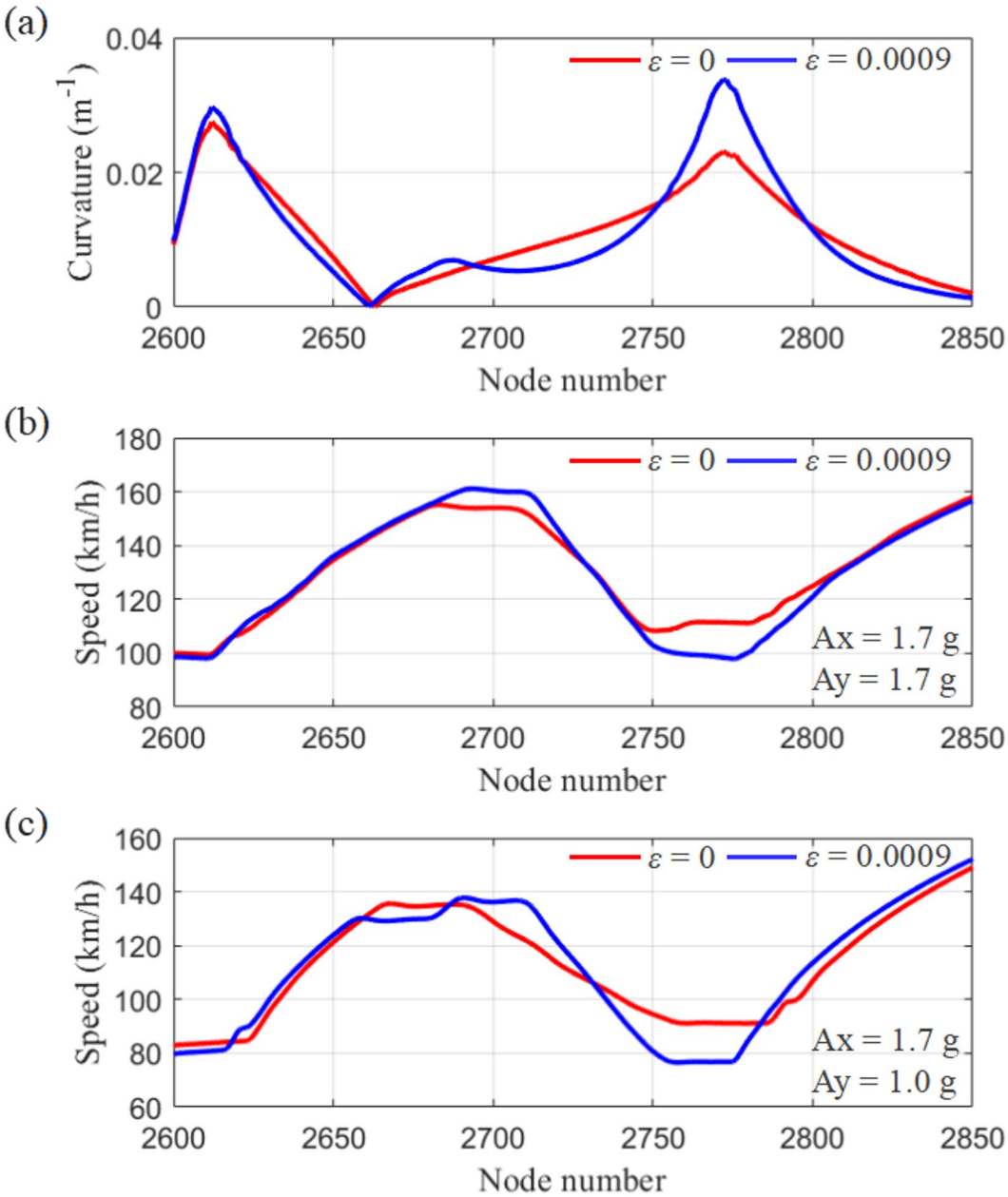

Fig. 11. Analysis about the reason why sometimes $\varepsilon = 0$ is faster but sometimes not. (**a**) comparison of curvature when $\varepsilon = 0.0009$ and $\varepsilon = 0$; (**b**) speed comparison when Ax = Ay = 1.7 g; (**c**) speed comparison when Ax = 1.7 g and Ay = 1.0 g.

Data availability

The datasets generated and analyzed during the current study are not publicly available due to secret geographic coordinates and vehicle parameters, but are available from the corresponding author (Dongmin Zhang) on reasonable request.

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Author contributions

Dongmin Zhang: Conceptualization, Methodology, Supervision, Review. Yi Yuan: Validation, Original draft, Visualization, Software, Validation.Futang Zhu: Project administration, Resources, Review.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to D.Z.

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